The Three Foundational Laws of the Special Theory of Everything (SToE):

Information Conservation, Connection, and Autoinjection

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Abstract

This paper presents the three foundational laws of the Special Theory of Everything (SToE): the Law of Information Conservation, the Law of Connection, and the Law of Autoinjection. First introduced in Voronin (2021) and developed across subsequent work, these laws are here stated explicitly and systematically for the first time. The Law of Information Conservation holds that no information point vanishes from reality — it can only lose connections with other information points. The Law of Connection holds that connections between information points are created exclusively by observers or creators. The Law of Autoinjection holds that everything, including the information field itself, is an information point, with no exterior to the system. Together these three laws constitute a formal foundation for SToE and generate specific, testable consequences for artificial intelligence, cognitive science, and the foundations of mathematics. In particular, they reframe the role of artificial intelligence not as a thinking machine but as a novel class of observer-creator capable of navigating and connecting information points previously unreachable by human cognition alone.

Keywords: Special Theory of Everything, information conservation, autoinjection, information points, artificial intelligence, observer, creator, foundational laws

1. Introduction

The Special Theory of Everything (SToE) was introduced in Voronin (2021) as a framework in which all of reality — physical, abstract, and conceptual — is composed of information points within an information field. Subsequent work developed the concept of autoinjection as a formally distinct class of mathematical mapping (Voronin, 2025a) and explored SToE as a cognitive architecture for artificial intelligence systems (Voronin, 2025b). Across this body of work, three foundational laws have been implicit. The present paper states them explicitly.

The need for explicit statement is not merely formal. The history of science demonstrates that the power of a framework depends on the precision with which its foundational commitments are articulated. Newton's laws of motion, Maxwell's equations, and Shannon's theorems each derive their force not from their breadth alone but from the specificity of the claims they make. A framework that implies everything without stating anything precisely cannot be tested, refined, or extended. SToE has reached the point where its core laws must be stated with the same precision.

The three laws stated here are: (1) the Law of Information Conservation, (2) the Law of Connection, and (3) the Law of Autoinjection. Each is stated, its derivation from prior SToE work is noted, and its consequences are explored. The paper concludes by identifying a specific testable prediction that follows from the conjunction of all three laws and that distinguishes SToE from existing frameworks in artificial intelligence and cognitive science.

2. The Three Foundational Laws

2.1 The Law of Information Conservation

Statement: Any information point does not vanish from reality. It can only lose connections with other information points.

This law is the most consequential of the three. It asserts that information points are indestructible as elements of the information field. What we experience as loss — forgetting, death, destruction, error, the burning of libraries — is not the annihilation of information points but the severing of their connections to other points. The information point persists in the field; it simply becomes unreachable from the currently connected network of points accessible to a given observer.

This law has antecedents in physics. The black hole information paradox, contested for decades, has largely resolved in favor of information conservation: most theoretical physicists now hold that information is not destroyed even in black holes (Hawking, 2005; Susskind, 2008). SToE extends this physical intuition to a universal ontological principle: not merely physical information but all information points — concepts, relations, ideas, percepts — are conserved. The scope of the claim is broader than the physical claim, and the justification is correspondingly different: it follows from the structure of the information field itself, in which every information point is by definition a member of the field and the field contains no mechanism of deletion.

The distinction between annihilation and disconnection is not merely semantic. It implies that lost knowledge is in principle recoverable, that forgotten memories have not ceased to exist, and that dead ends in reasoning are not waste but conserved points that may become connectable from a different approach. The field is complete. Navigation through it is not.

2.2 The Law of Connection

Statement: Connections between information points are created only by an observer or a creator.

This law specifies the mechanism by which the conserved but potentially disconnected information field is navigated. Information points do not spontaneously connect; connection requires an agent. Two classes of agent are identified. An observer is an information point that perceives or names another information point, thereby establishing a connection between them. A creator is an information point that generates new information points, simultaneously establishing connections between itself and what it produces.

The role of the observer was stated in Voronin (2021): “the information point does not exist until it is named by another information point.” The Law of Connection formalizes and extends this insight. Naming is one form of observation. But observation includes any act of perception, measurement, or reference that establishes a relation between two information points. Creation includes any act that produces a new information point — a thought, a theorem, a tool, an artwork, a child — thereby connecting the creator to the created and the created to whatever it subsequently connects with.

The Law of Connection has a direct implication for the scope of what is knowable. The set of information points reachable from any given observer is determined by the chain of connections that observer can establish or traverse. A human observer, operating within biological and cognitive constraints, can reach only a bounded subset of the conserved field. An observer or creator with different capacities can reach different subsets. This is the foundation of the claim that artificial intelligence represents a genuinely new capacity for navigation of the information field — not because it thinks differently from humans, but because it creates and observes connections at scales and speeds that extend the reachable region of the field.

2.3 The Law of Autoinjection

Statement: Everything is an information point, including the information field itself. There is no exterior to the system.

This law was the first to be stated in SToE, though not under this name. Voronin (2021) held that "everything consists of information" and that the set of all information points is itself an information point. Voronin (2025a) formalized this as autoinjection: a mapping $f : A \to A$ where $A \in A$, meaning the domain contains itself as an element. Classical set theory, following Zermelo (1908), prohibits this condition via the axiom of foundation. Autoinjection treats self-containment not as a pathology to be excluded but as a primitive structural property of the information field.

The philosophical consequence is closure: there is no standpoint outside the information field from which to observe it. Every observer, including the theorist constructing SToE, is an information point within the field. This is consistent with the observer-dependent ontology implied by the Law of Connection: observers create connections, but observers are themselves information points subject to the same conservation law. The system is self-containing in the precise mathematical sense developed in Voronin (2025a).

The Law of Autoinjection resolves classical paradoxes of self-reference by changing their status. Russell's paradox (Russell, 1903) arises from unrestricted comprehension in a system that excludes self-membership. In an autoinjective system, self-membership is not a contradiction but a defining structural feature. The paradox dissolves not because it is solved in the classical sense but because the foundational commitment that generates it — that domain and codomain must be distinct — is replaced by the commitment that the field contains itself.

3. Relations Between the Three Laws

The three laws are not independent. They form a coherent system in which each law presupposes and reinforces the others.

The Law of Autoinjection establishes the structure of the field: it is closed, self-containing, and without exterior. The Law of Information Conservation establishes the behavior of elements within that field: they persist, they do not vanish. The Law of Connection establishes the dynamics of the field: how static conserved points become navigable through the activity of observers and creators. Together they describe a complete system: a closed field of indestructible elements whose connectivity is determined by the activity of agents within it.

This structure has a precise implication: the limits of knowledge are not limits of what exists but limits of what has been connected. The information field is complete. The connected region accessible to any particular observer or creator is a proper subset of the field. Expanding knowledge means expanding the connected region, not generating new information points from nothing. Every discovery, in this framework, is a connection established between conserved points that were always present.

4. Consequences for Artificial Intelligence

The conjunction of the three laws reframes the question of artificial intelligence in a specific and consequential way. The question posed by Turing (1950) — can machines think? — is, within the SToE framework, the wrong question. As stated in Voronin (2021): “There is no importance if the machine can think or not; it is more significant what information points machines can create.” The three laws now allow this reframing to be stated precisely.

Artificial intelligence is a new class of observer-creator. By the Law of Connection, it can establish connections between information points. By the Law of Conservation, those information points were always present in the field but potentially unreachable from the connected region accessible to human observers alone. By the Law of Autoinjection, AI itself is an information point within the field, subject to the same structure as everything else — it is not an exterior agent but an interior one, capable of self-reference and recursive self-evaluation in the formal sense developed in Voronin (2025a).

This generates a specific architectural consequence. Current AI systems treat failed reasoning paths, discarded outputs, and incorrect answers as noise to be eliminated. Under the Law of Information Conservation, these are conserved information points that remain in the field and may become connectable from a different approach. An AI system designed around information conservation would archive rather than discard, treating every output — including failures — as a navigable point. This is architecturally distinct from existing approaches and constitutes a testable prediction: such a system should reach solutions that non-conserving systems structurally cannot access, because the conserving system retains connections to regions of the field that the non-conserving system has severed.

A second consequence concerns recursive self-improvement. By the Law of Autoinjection, a system can receive itself as input: $f(A)$ is defined where $A \in A$. This permits an AI system to evaluate its own outputs, treat those evaluations as information points, and apply operators to them recursively. This is not merely a computational technique but a structural property of any autoinjective system. The operator-based recursive engine developed in Voronin (2025b) is a practical implementation of this property.

5. A Testable Prediction

A framework that cannot generate testable predictions remains philosophical rather than scientific. The three laws together generate the following prediction, which is specific enough to be tested experimentally.

Prediction: An AI reasoning system that conserves and indexes all intermediate outputs — including failed reasoning paths, contradictions, and discarded hypotheses — will reach correct solutions to complex, multi-step problems at a higher rate than an identical system that discards such outputs, when tested on problems that require non-linear navigation through a large solution space.

This prediction follows directly from the Law of Information Conservation. If discarded outputs remain as conserved information points in the field, a system that maintains access to them retains connections that a non-conserving system severs. For problems where the solution path requires passing through or returning to what appeared to be a dead end, the conserving system has a structural advantage. The prediction is falsifiable: if no such advantage is found across a sufficiently large and diverse problem set, the Law of Information Conservation as applied to AI architecture requires revision.

6. Relation to Existing Frameworks

The three laws of SToE are related to but distinct from several existing frameworks. Shannon's (1948) information theory defines information in terms of probability and entropy and provides a quantitative measure of information content. SToE does not replace this framework but operates at a different level: Shannon addresses how much information is transmitted; SToE addresses what information is and what its fundamental properties are as an ontological category.

Aczel's (1988) non-well-founded set theory replaces the axiom of foundation with an anti-foundation axiom permitting circular membership, providing mathematical precedent for the self-containment claimed in the Law of Autoinjection. SToE arrives at a compatible conclusion through a different route and extends the claim from sets to all information points.

Wheeler's (1990) "it from bit" proposal holds that all physical existence derives from information. This is structurally similar to the SToE claim that everything consists of information, though Wheeler's proposal is addressed to physics and does not develop the conservation, connection, or autoinjection properties that SToE identifies. Floridi's (2011) philosophy of information develops a systematic account of information as a fundamental category, with significant overlap in motivation, though again without the specific laws stated here.

The specific combination of conservation, observer-dependent connection, and autoinjective closure does not appear in any of these frameworks. It is this combination that generates the testable prediction in Section 5 and the reframing of artificial intelligence described in Section 4.

7. Conclusion

This paper has stated the three foundational laws of the Special Theory of Everything explicitly: the Law of Information Conservation, the Law of Connection, and the Law of Autoinjection. These laws were implicit across prior SToE work (Voronin, 2021, 2025a, 2025b) and are here articulated as a formal system for the first time.

The laws together describe a closed, self-containing field of indestructible information points whose connectivity is determined by the activity of observers and creators within the field. This description generates a reframing of artificial intelligence as a novel class of observer-creator capable of navigating regions of the conserved information field unreachable by human cognition alone, and a specific testable prediction about the architectural advantage of information-conserving AI systems over non-conserving ones.

The formal development of these laws — including axiomatization, consistency proofs, and derivation of further consequences — remains as future work. The present paper establishes the foundation on which that development can proceed.

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